

Analysis Note:
 $B^0 \rightarrow K^{*0} \ell^+ \ell^-$ Sensitivity with an AI-Agent-Driven,
Open-Source Fast-Simulation Framework

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with an LLM analysis agent (Claude) operating under human review

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This study uses only publicly available tools and publications. No Belle II internal materials
— data, MC, internal notes, or the software framework — are used. Everything is
open-source and runs on a laptop.
Companion to the $\mathcal{B}(B^0 \rightarrow D^{*-} \tau^+ \nu)$ note, which documents the framework in full.

Abstract

We apply the AI-agent analysis framework to a topology complementary to the semileptonic worked example: the fully reconstructed rare decay $B^0 \rightarrow K^{*0} \ell^+ \ell^-$ ($\ell = e, \mu$), where the discriminating variables are the full-reconstruction pair $M_{bc}/\Delta E$ and the dominant background is the charmonium peaking contribution $B^0 \rightarrow J/\psi(\rightarrow \ell\ell) K^{*0}$ at $\sim 10^3$ times the signal rate. With an $M_{bc}/\Delta E$ selection and per-flavor asymmetric charmonium vetoes — the wider electron windows being required by the radiative tail the simulation reproduces — the expected statistical precision on $\mathcal{B}(B^0 \rightarrow K^{*0} \ell^+ \ell^-)$ at 1 ab^{-1} is 8–10% per lepton channel, with the $J/\psi K^*$ background suppressed from $\sim 25\,000$ to a few hundred events. Combinatorial backgrounds are not yet modeled; the quoted precision is therefore a best case, and the corresponding data-driven strategy is outlined.

1 Introduction

The electroweak-penguin decay $B^0 \rightarrow K^{*0} \ell^+ \ell^-$ proceeds through a $b \rightarrow s \ell \ell$ flavor-changing neutral current, with $\mathcal{B} \approx 1 \times 10^{-6}$ [1]; it is a standard probe of physics beyond the Standard Model. Our purpose here is narrower: to show that the same agent-driven pipeline documented in the companion $B^0 \rightarrow D^{*-} \tau^+ \nu$ note handles a *fully reconstructed* topology, where the analysis variables and background structure are entirely different — M_{bc} and ΔE instead of missing mass, and a peaking charmonium background instead of combinatorial lepton pairing. The selection strategy follows the $B \rightarrow K^{(*)} \ell \ell$ measurements of Belle and BaBar [2, 3]: $M_{bc}/\Delta E$ signal windows and vetoes in the dilepton mass around the J/ψ and $\psi(2S)$.

2 Samples

Generation follows the companion note (EvtGen with Photos, Belle II-like boost, forced chains, fixed seeds; detector response and lepton identification identical, Table 2 there).

The signal model is the BTOSLLBALL form-factor implementation of the $b \rightarrow s \ell \ell$ effective Hamiltonian; the $J/\psi K^*$ sample uses the measured transversity amplitudes. Candidates are built from the true $K^+ \pi^-$ pair and the two leptons, all passed through the detector response; both leptons

Table 1: Samples for this study.

Process	Model	Events	Seeds	Role
$B^0 \rightarrow K^{*0}(\rightarrow K^+\pi^-)\ell^+\ell^-$	BTOSLLBALL [4]	10 000	32/33	signal ($ee:\mu\mu = 1:1$)
$B^0 \rightarrow J/\psi(\rightarrow \ell\ell)K^{*0}$	SVV_HELAMP + VLL	5 000	34/35	peaking background

must pass lepton identification ($\varepsilon_e = 95\%$, $\varepsilon_\mu = 90\%$), and identified tracks receive the corresponding mass hypothesis. Candidate-level variables are the standard

$$M_{bc} = \sqrt{(\sqrt{s}/2)^2 - |\vec{p}_B^{\text{CM}}|^2}, \quad \Delta E = E_B^{\text{CM}} - \sqrt{s}/2, \quad m(\ell^+\ell^-),$$

the basf2 `Mbc/deltaE` definitions.

3 Selection

Figure 1 shows M_{bc} and ΔE for the signal by lepton flavor. M_{bc} peaks at m_B with $\sigma \approx 15$ MeV; ΔE peaks near zero with an asymmetric radiative tail (mean -25 MeV, RMS 137 MeV), more pronounced for electrons. The selection is

$$|M_{bc} - m_B| < 30 \text{ MeV}, \quad -0.30 < \Delta E < 0.25 \text{ GeV}.$$

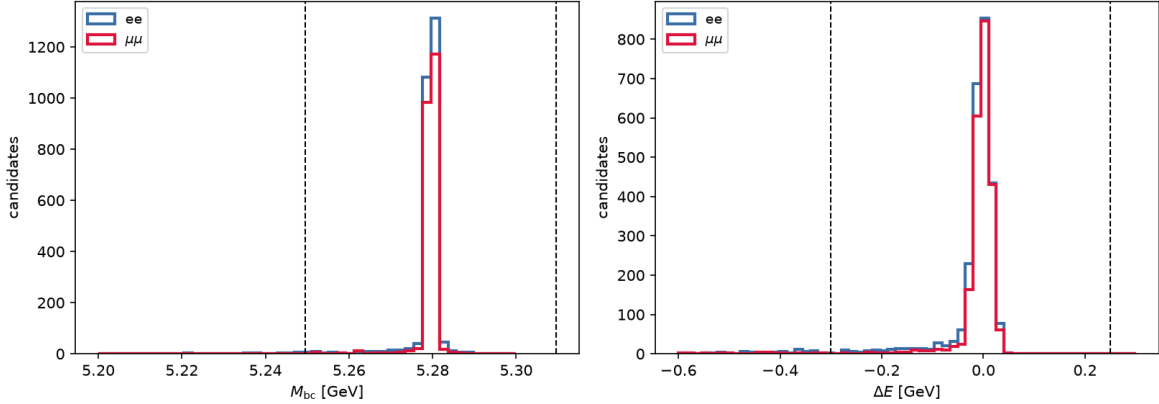


Figure 1: Signal M_{bc} (left) and ΔE (right) by lepton flavor, with the selection indicated.

4 Background: the charmonium veto

$B^0 \rightarrow J/\psi(\rightarrow \ell\ell)K^{*0}$ and $\psi(2S)K^{*0}$ have exactly the signal's visible final state and peak identically in $M_{bc}/\Delta E$, at $\mathcal{B}(B^0 \rightarrow J/\psi K^{*0}) \times \mathcal{B}(J/\psi \rightarrow \ell\ell) \approx 7.6 \times 10^{-5}$ per channel — roughly $70\times$ the signal. They are removed with windows in $m(\ell^+\ell^-)$ (Fig. 2, left). Radiation shifts $m(\ell^+\ell^-)$ below the resonance, and more strongly for electrons; the vetoes are therefore asymmetric and flavor-dependent, exactly as in the published analyses [2]:

$$ee : m(\ell^+\ell^-) \notin (2.85, 3.20) \cup (3.54, 3.77) \text{ GeV}; \quad \mu\mu : m(\ell^+\ell^-) \notin (2.98, 3.18) \cup (3.59, 3.77) \text{ GeV}.$$

Backgrounds *not* modeled in this study, with the corresponding data strategy:

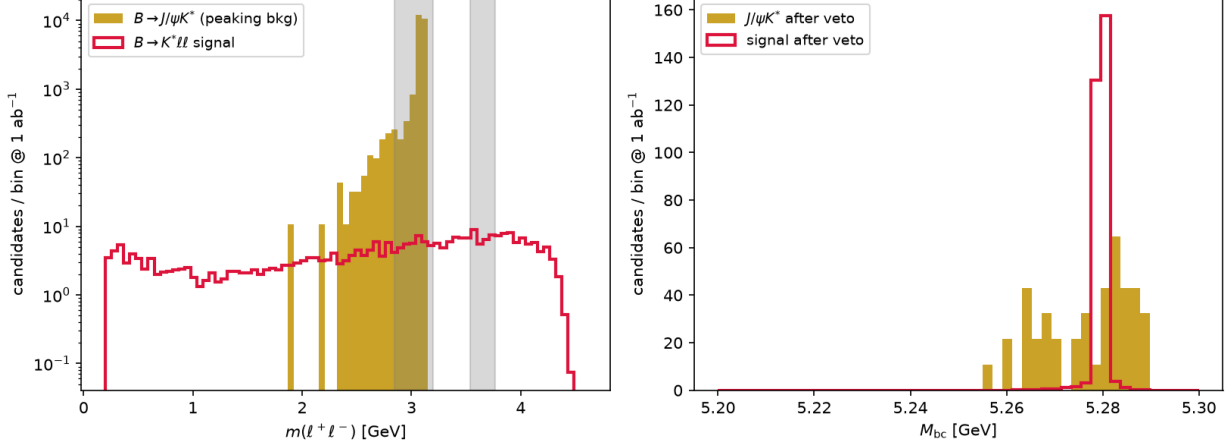


Figure 2: Left: $m(\ell^+\ell^-)$ for signal and $J/\psi K^*$ normalized to 1 ab^{-1} , with the electron veto windows shaded. Right: M_{bc} after the veto.

- **Combinatorial** (generic $B\bar{B}$ and continuum with two real or fake leptons): flat in M_{bc} ; on data it is normalized in the M_{bc} sideband ($M_{bc} < 5.26 \text{ GeV}$), the standard procedure. Its omission is why the precision below is a best case.
- **Double hadron misidentification** ($B^0 \rightarrow K^{*0}\pi^+\pi^-$ with both pions faking leptons): suppressed by the fake-rate squared; with the framework's rates and $\mathcal{B}(B^0 \rightarrow K^{*0}\pi^+\pi^-) \approx 5.5 \times 10^{-5}$ this is $\lesssim 2\%$ of the signal and currently negligible.
- The vetoed J/ψ sample itself is the standard *control channel* on data: with $\sim 25\,000$ events per ab^{-1} it calibrates the $M_{bc}/\Delta E$ shapes, the lepton-ID efficiencies, and the tail modeling with sub-percent statistical precision.

5 Results

Table 2: Efficiencies and expected yields at 1 ab^{-1} ($N_{B^0\bar{B}^0} = 2f^{00}\sigma_{B\bar{B}}L$, single $K^{*0} \rightarrow K^+\pi^-$ chain). The precision estimate takes the residual $J/\psi K^*$ leakage as the only background.

Channel	ε	S	$J/\psi K^*$ before veto	after veto	$\delta\mathcal{B}/\mathcal{B}$ (stat)
ee	43.1%	317	25 682	368	8.3%
$\mu\mu$	38.2%	286	23 070	518	9.9%

The veto suppresses the peaking background by a factor 50–70 at a signal cost of $\lesssim 10\%$. The residual leakage enters through the resolution and radiative tails; on data it would be constrained by the J/ψ control channel. No fit is performed at this stage; the natural next step is a one-dimensional M_{bc} fit (signal peak + ARGUS-shaped combinatorial + peaking residual), which requires the combinatorial samples first.

6 Discussion and conclusion

This study demonstrates that the framework’s reach is not limited to missing-energy topologies: switching from $B^0 \rightarrow D^* \tau \nu$ to $B^0 \rightarrow K^{*0} \ell^+ \ell^-$ changed the decay file, the candidate topology, and the analysis variables, but no framework code. Two findings are worth noting. First, the simulation reproduces the radiative dilepton-mass tail that forces flavor-dependent asymmetric charmonium vetoes — a nontrivial qualitative agreement with real-analysis practice. Second, the limitations are complementary to the semileptonic note: here the missing ingredient is the combinatorial background (generic $B\bar{B}$ /continuum with a true K^* and two leptons, and fake- K^* combinations), which requires extending the candidate finder beyond truth seeding; this is the same upgrade the semileptonic analysis needs for its Δm sideband, and it is the top-priority framework item.

At 1 ab^{-1} the expected statistical precision is 8–10% per channel under the stated assumptions — sufficient for a meaningful branching-fraction measurement, and the starting point for the angular observables that make this channel interesting once a fit framework for them exists.

References

- [1] S. Navas *et al.* [Particle Data Group], Phys. Rev. D **110**, 030001 (2024).
- [2] J.-T. Wei *et al.* [Belle], “Measurement of the Differential Branching Fraction and Forward-Backward Asymmetry for $B \rightarrow K^{(*)} \ell^+ \ell^-$,” Phys. Rev. Lett. **103**, 171801 (2009).
- [3] J. P. Lees *et al.* [BaBar], “Measurement of Branching Fractions and Rate Asymmetries in the Rare Decays $B \rightarrow K^{(*)} \ell^+ \ell^-$,” Phys. Rev. D **86**, 032012 (2012).
- [4] A. Ali, P. Ball, L. T. Handoko, G. Hiller, “A comparative study of the decays $B \rightarrow (K, K^*) \ell^+ \ell^-$ in standard model and supersymmetric theories,” Phys. Rev. D **61**, 074024 (2000).